

Khushi Singh

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EDUCATION

Vivekanand Education Society's Institute of Technology

Bachelor of Engineering – Information Technology CGPA: 9.92 / 10 (till Sem 6)

Mumbai, India

2023 - 2027

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript, C, Bash, Solidity

Frameworks: REST APIs, FastAPI, Node.js, React

Libraries & ML: TensorFlow, HuggingFace, XGBoost, LangChain

Tools & Other: Git, MySQL, Docker, MongoDB, Firebase, Blockchain (Polygon PoS, Solidity), Jira, Postman, CI/CD (Jenkins), GitHub Copilot

EXPERIENCE

Veermata Jijabai Technological Institute (VJTI)

Mumbai, India

Software Developer & Research Intern | Python, TensorFlow, HuggingFace, Medical Imaging

May 2025 - Jul 2025

- Developed deep learning models for automated disease classification from medical scans (MRIs, CT, X-rays) using TensorFlow and HuggingFace; applied CNN architectures (ResNet) and vision transformers, evaluating on precision, recall, and AUC-ROC metrics.
- Built end-to-end Python preprocessing pipelines for medical image data augmentation, normalisation, and class-imbalance correction to improve model generalisation across multi-class clinical datasets.

ISTE-VESIT

Mumbai, India

Junior Public Relations Officer

Aug 2024 - June 2026

- Managed communications and stakeholder coordination for **10+** technical events per year, driving a **30%** increase in student participation.

PROJECTS

BioToken: Reagent Provenance & Verification | *Python, XGBoost, Solidity, Polygon PoS, ZK-SNARKs, FastAPI* | [GitHub](#)

- Built an end-to-end reagent authentication pipeline from manufacturer HPLC scan to on-chain ZK proof addressing supply chain fraud without exposing proprietary chemical data, at **\$0.0087 per vial** on Polygon PoS.
- Trained an XGBoost anomaly classifier on **77,901 molecules** (137 RDKit features) to detect tampered reagents via HPLC retention-time deviation; achieved **AUC 0.9798, F1 0.9351** validated on **1,000 real multi-lab entries** and **15 published degradation measurements**.
- Deployed **2 Solidity smart contracts** managing **5 lifecycle states** with role-based access control; FastAPI backend handles webhook-based custody logging and a cross-company consensus validator.

RecBlock: Web 2.5 Decentralised EHR System | *Solidity, Polygon PoS, IPFS, React, Node.js, AES-256* | [GitHub](#)

- Built **RecBlock** as a Web 2.5 bridge giving patients full cryptographic record ownership with **Web2-grade usability**, eliminating wallet and gas-fee friction via **ERC-4337 Account Abstraction** (Privy + Pimlico Paymaster) at **INR 0.27 (\$0.003) per transaction**.
- Achieved **1.5s average retrieval** by storing AES-256-GCM encrypted records on IPFS and decrypting locally, bypassing on-chain consensus for reads while anchoring **keccak256 integrity proofs** on Polygon PoS for tamper-proof audit trails.

NaviSense: Inclusive Visual Assistance Platform | *Flutter, YOLOv8n TFLite, Python, Firebase*

- Deployed a **quantized YOLOv8n TFLite** model entirely on-device with **zero network dependency**, achieving real-time inference at **1 frame/3s** for **70M+** visually impaired users in low-connectivity regions.
- Won 1st Place, Hack4Innovation** (VESIT × Rotary Club of Mumbai, Mar 2026)

CERTIFICATIONS

Oracle Academy - Database Design (90 hrs, 96%), 2025-26

Oracle Academy - Database Programming with SQL (90 hrs, 82%), 2025-26

AWS Academy Graduate - Cloud Foundations - Amazon Web Services, October 2025

MathWorks Onramp Series - MATLAB, Machine Learning & Deep Learning - MathWorks, February 2026

ACHIEVEMENTS

Winner, Hack4Innovation (VESIT × Rotary Club of Mumbai), March 2026.

Presented Research Paper, "BioToken: Decentralized Reagent Provenance & Verification using ZK-Proofs and Blockchain," IJCACI 2026, WUST, USA, Jul 2026 (Springer, in press).

3rd Prize, Manthan: Deep Thinking - Aarohan 2026, VESIT Department Technology Day, March 2026.

2nd Runner-Up, Genesis Hackathon - AI/ML and Blockchain track, among 500+ registrations.

2nd Position, Youth Conclave, Sydenham College of Commerce and Economics, Mumbai, 2025.